OpenAI: Prompt Engineering for IT Administrators

Prompting ChatGPT for Maximum Effectiveness

Tim Warner
Principal Author, Pluralsight
@TechTrainerTim | TechTrainerTim.com

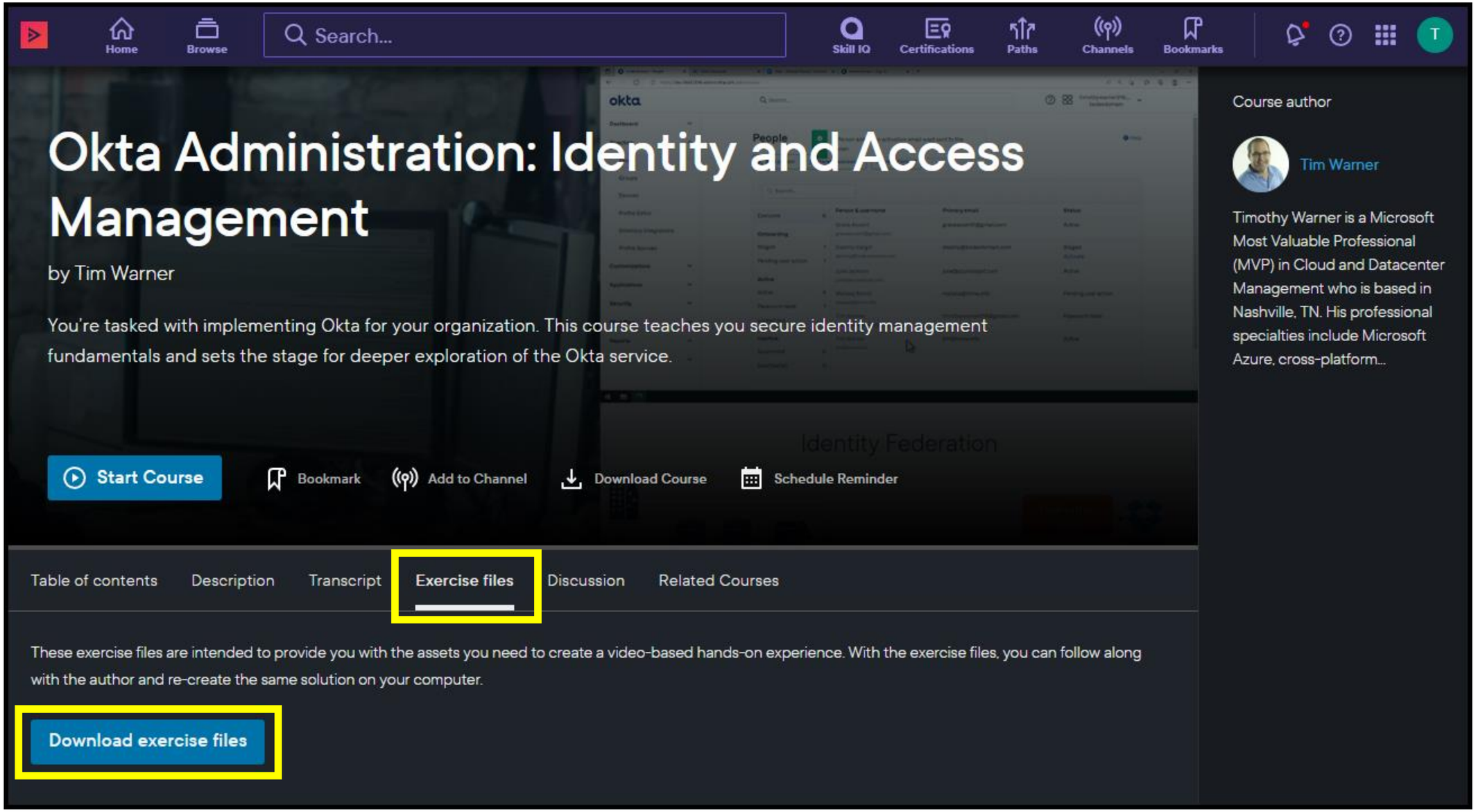
Overview

Review prompt engineering basics

Demonstrate effective prompts applied to everyday IT administration tasks

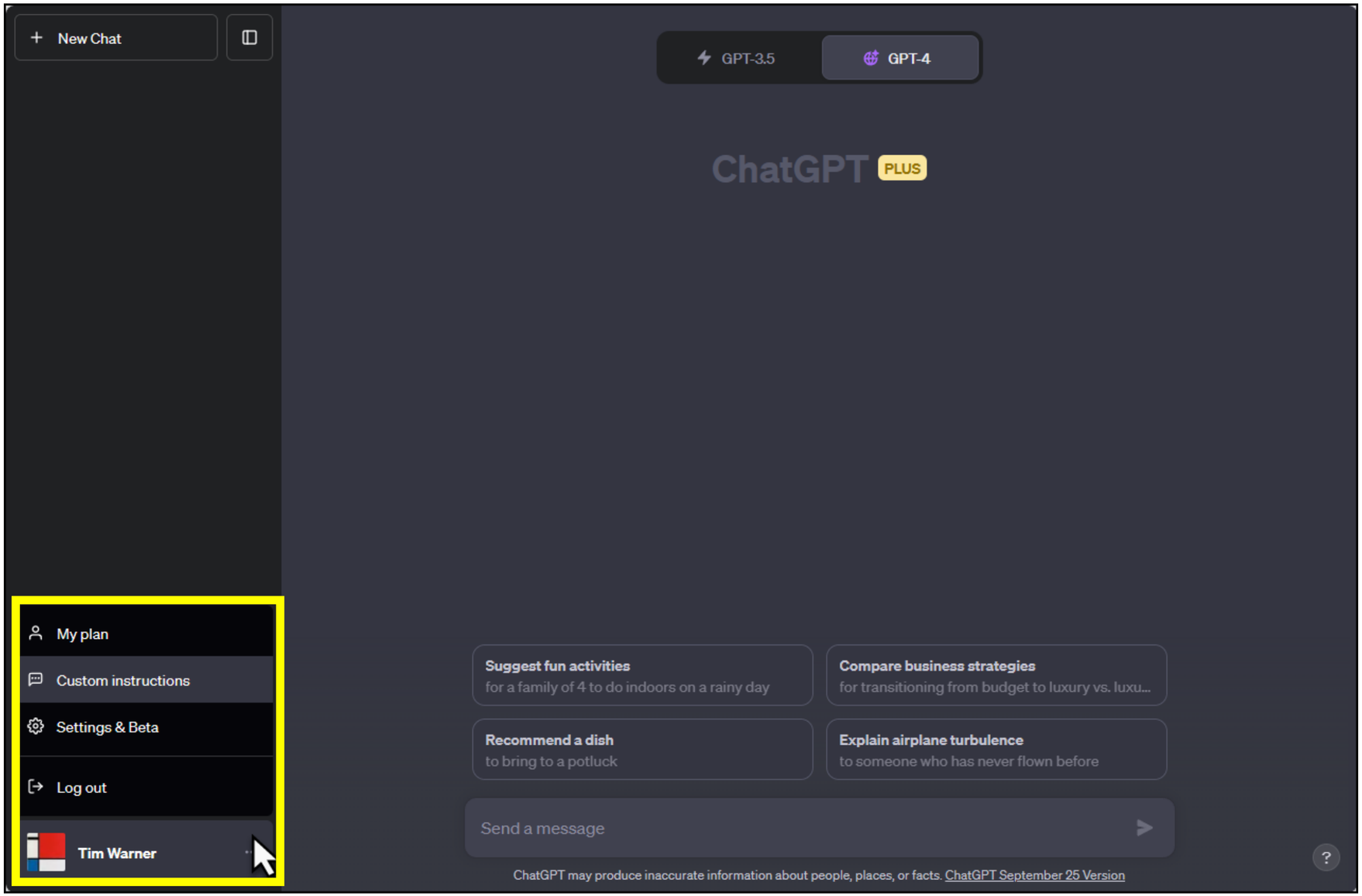
Case study scenario format

Exercise Files

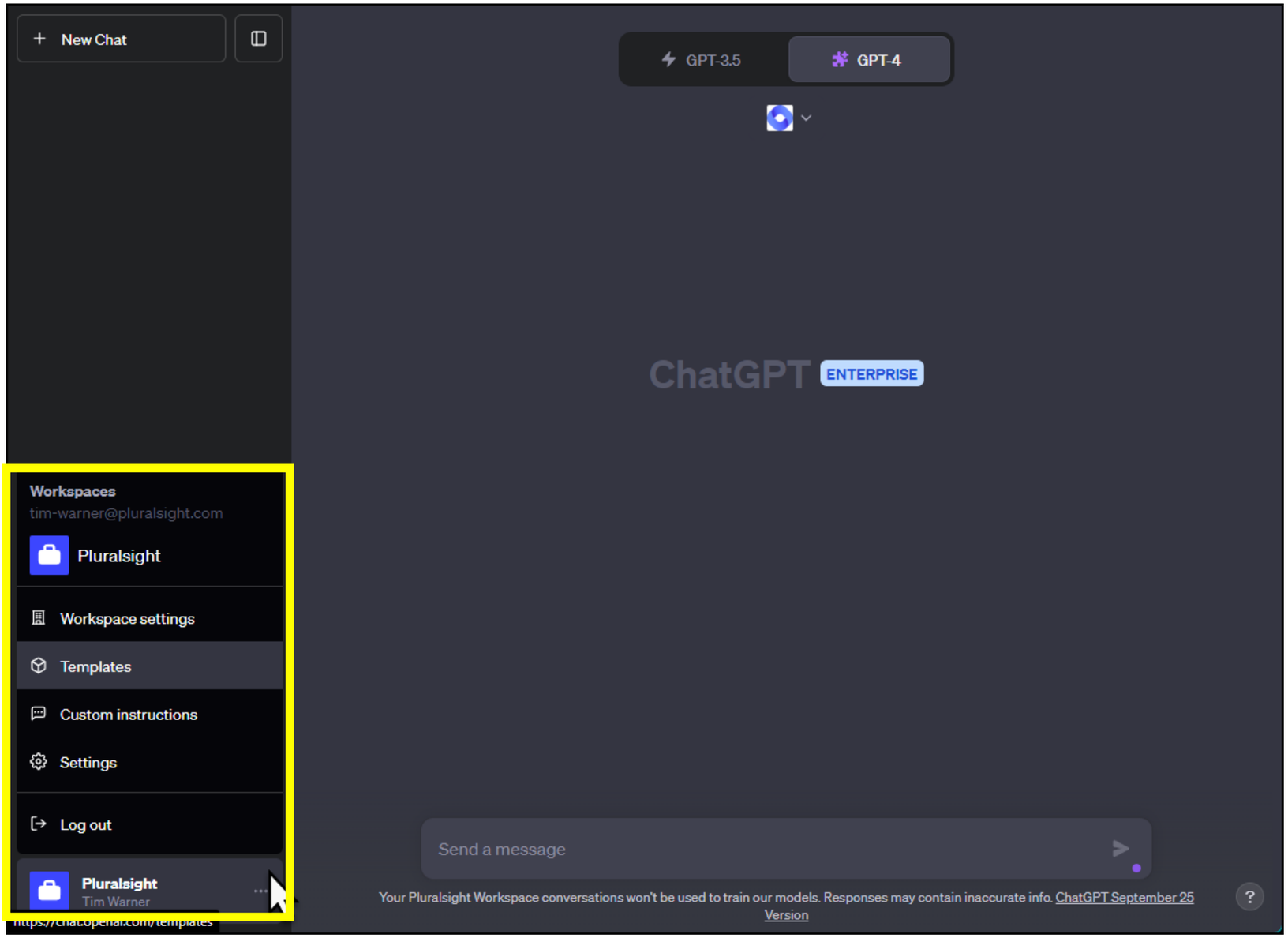


Prompt Engineering Basics

ChatGPT UI


New Chat
GPT-3.5
GPT-4
ChatGPT PLUS
My plan
Custom instructions
Settings & Beta
Log out
Tim Warner
Suggest fun activities
for a family of 4 to do indoors on a rainy day
Compare business strategies
for transitioning from budget to luxury vs. luxu...
Recommend a dish
to bring to a potluck
Explain airplane turbulence
to someone who has never flown before
Send a message
ChatGPT may produce inaccurate information about people, places, or facts. ChatGPT September 25 Version

ChatGPT Enterprise

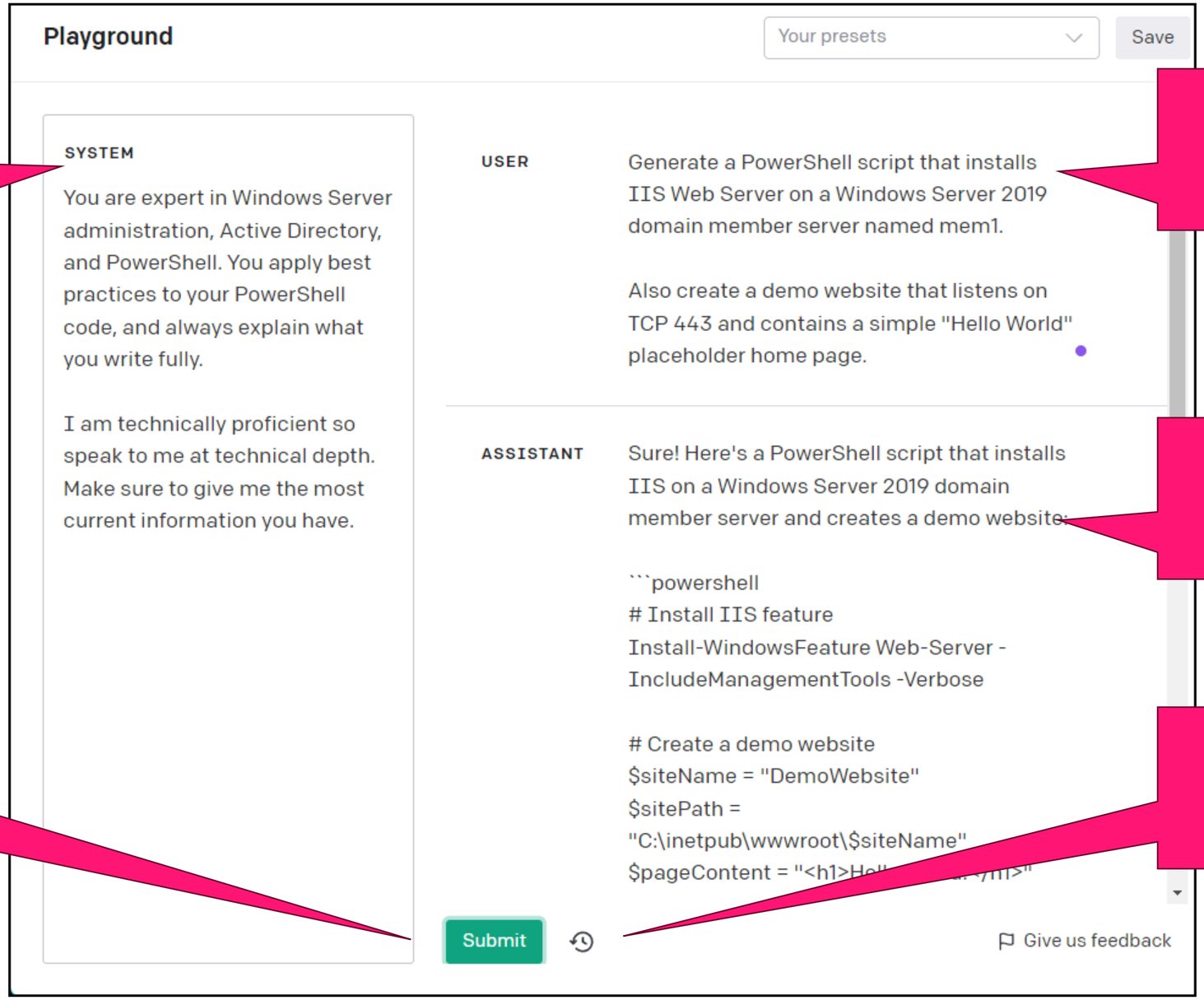


Prompt engineering

The process of crafting and refining prompts (inputs) to guide the response of a Generative Pre-Trained Transformer (GPT) model towards desired outputs or behaviors.

Citation: ChatGPT

Anatomy of a ChatGPT Conversation



System prompt

User prompt

Completion

Max token length

Statefully stateless

GPT Prompting Proven Practices

Explicit instruction

- Keep (max) token length in mind
- Zero-shot learning

Priming

- Provide model with examples to guide its behavior
- Few-shot learning

Persona modeling

- Assign a character to the language model

Iteration

- Refining your prompt through a multi-turn conversation
- OpenAI Playground

Scenarios

Case Study

GLOBOMANTICS Academy

Independent high school

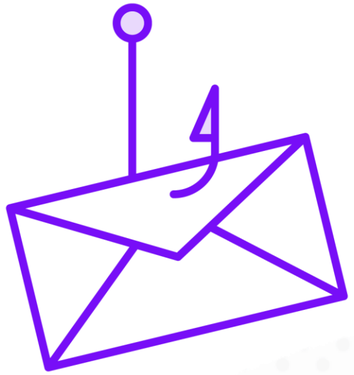
500 students, 125 faculty and staff

Microsoft Active Directory back-end with Microsoft Azure and Microsoft 365 service integration

Students and faculty use macOS; staff and IT use Windows

(Very) small IT shop - you wear several hats

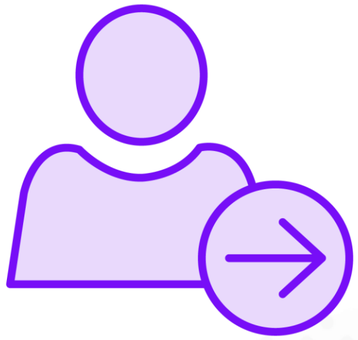
Scenario 1: Phishing User Education



You need to educate your users on how to avoid phishing attacks via an email "blast"

Problem: You have little aptitude for crafting language, business or otherwise

Scenario 2: Bulk-Create User Accounts



You need to create 125 Active Directory user accounts for the next school term

Problem 1: You don't have the time or willingness to undertake this task manually

Problem 2: You've put off learning PowerShell for way too long

Demo

user data analysis to make student CSV

custom instructions

prompt templates

Summary

Prompting is how you "program" the GPT model

Might be worth your while to study technical writing to improve your prompts

Thanks so much!

Courses: timw.info/ps

Email: tim-warner@pluralsight.com